arXiv Preprint

arxiv.org/abs/2607.07780

Colimits of Enriched Categories

Andy Sukowski-Bang Kiel University uni@andy.sb andy.sb

Abstract. We introduce categories of strings to give an explicit hom-object formula for colimits of enriched categories. As corollaries, we recover the rigidification of simplicial sets via Dugger and Spivak's necklace construction and describe colimits in Cat and of Lawvere metric spaces.

MathOverflow Summary

mathoverflow.net/a/513164

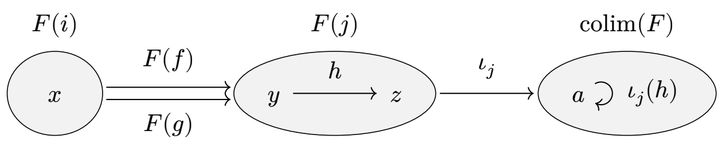
Colimits in Cat

Coequalizer Example in Cat

Consider the category $\mathcal{I} := \{f, g\colon i \rightrightarrows j\}$ and the following diagram $F\colon \mathcal{I} \to \mathrm{Cat}$ where

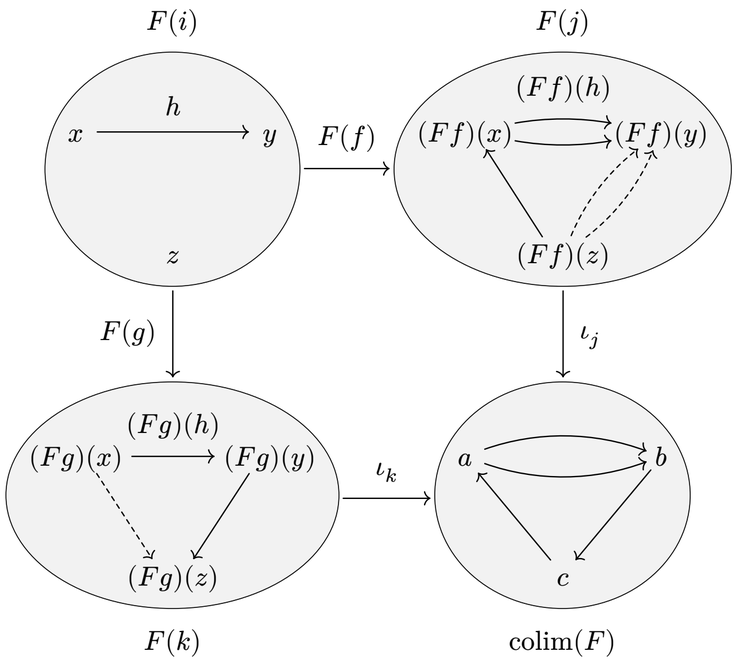
$$y = (Ff)(x), \qquad z = (Fg)(x).$$

Then the coequalizer $\mathrm{colim}(F)$ with colimiting cocone $\iota\colon F \Rightarrow \Delta_{\mathcal{I}}(\mathrm{colim}\, F)$ has a single object with infinitely many endomorphisms.



Pushout Example in Cat

Let $\mathcal{I}$ be the span $k \xleftarrow{g} i \xrightarrow{f} j$ and $F\colon \mathcal{I} \to \mathrm{Cat}$ the diagram below, where dashed arrows indicate compositions. The pushout $\mathrm{colim}(F)$ with colimiting cocone $\iota\colon F \Rightarrow \Delta_{\mathcal{I}}(\mathrm{colim}\, F)$ has objects $a = \iota_i(x)$, $b = \iota_i(y)$ and $c = \iota_i(z)$.



Some hom-sets are "glued together". The indicated morphisms in $\mathrm{colim}(F)$ freely generate infinitely many compositions.

Colimits in Cat

Let $F\colon \mathcal{I} \to \mathrm{Cat}$ be small and let $a, b \in \mathrm{colim}(\mathrm{Ob} \circ F)$. Then each morphism in $(\mathrm{colim}\, F)(a, b)$ is an equivalence class of strings $[(i_n, \alpha_n), ..., (i_1, \alpha_1)]$ where

$$\alpha_k \in \mathrm{mor}\, F(i_k), \qquad \mathrm{dom}(\alpha_1) \in a,$$
$$\mathrm{cod}(\alpha_{k-1}) \sim \mathrm{dom}(\alpha_k), \qquad \mathrm{cod}(\alpha_n) \in b.$$

The equivalence relation of strings is generated by compositions of consecutive morphisms in the same $F(i)$ and edgewise transports along morphisms $f\colon i \to j$ in $\mathcal{I}$.

$$[..., (i, \beta), (i, \alpha), ...] \sim [..., (i, \beta \circ \alpha), ...]$$
$$[..., (i, \alpha), ...] \sim [..., (j, (Ff)(\alpha)), ...]$$

[Lur08] J. Lurie, *Higher Topos Theory*. 2008.

[DS11] D. Dugger and D. I. Spivak, "Rigidification of quasi-categories," 2011.

Colimits in $\mathcal{V}$-Cat

Hom-Object Formula

Let $\mathcal{V}$ be cocomplete and locally small, and let $\otimes$ be cocontinuous in each argument. Let $F\colon \mathcal{I} \to \mathcal{V}\text{-Cat}$ be small.

$$\mathrm{Ob}(\mathrm{colim}\, F) \cong \mathrm{colim}(\mathrm{Ob} \circ F).$$

We will introduce **categories of strings** $\mathcal{S}(a, b)$ and **string evaluation** functors $\mathrm{ev}_{a,b}\colon \mathcal{S}(a, b) \to \mathcal{V}$, such that $\mathrm{colim}(F)$ has hom-objects

$$(\mathrm{colim}\, F)(a, b) \cong \operatorname*{colim}_{\mathcal{S}(a,b)} \big(\mathrm{ev}_{a,b}\big).$$

Categories of Strings

Let $F\colon \mathcal{I} \to \mathcal{V}\text{-Cat}$ be small. The category of elements

$$\int_{i \in \mathcal{I}} \mathrm{Ob}(Fi) \times \mathrm{Ob}(Fi)$$

has as objects **edges** (i, x, y) with $x, y \in F(i)$ and morphisms $(i, x, y) \to (i', x', y')$ along $f\colon i \to i'$ in $\mathcal{I}$, such that $(Ff)(x) = x'$ and $(Ff)(y) = y'$.

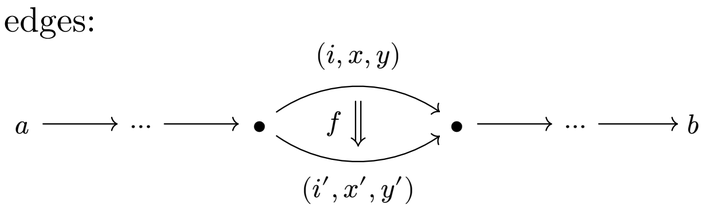
These edges constitute a directed graph with vertices $\mathrm{colim}(\mathrm{Ob} \circ F)$, where (i, x, y) is a directed edge from $[x]$ to $[y]$.

$$[x] \xrightarrow{(i_1, x_1, y_1)} [y]$$

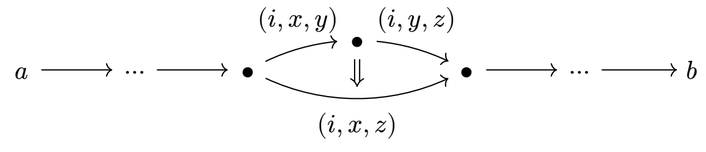
Let $a, b \in \mathrm{colim}(\mathrm{Ob} \circ F)$. Define the **category of non-empty strings** $\mathcal{S}(a, b)$, whose objects are paths in said graph, which we call **strings** from a to b.

$$a \xrightarrow{(i_1, x_1, y_1)} \bullet \xrightarrow{(i_2, x_2, y_2)} \cdots \xrightarrow{(i_n, x_n, y_n)} b \qquad (1)$$

Morphisms in $\mathcal{S}(a, b)$ are generated by **transports** along underlying morphisms f between edges:



and **contractions** of adjacent edges with the same $i \in \mathcal{I}$:



Evaluation of Strings

Let $F\colon \mathcal{I} \to \mathcal{V}\text{-Cat}$ be small. Define the **string evaluation** functor $\mathrm{ev}_{a,b}\colon \mathcal{S}(a, b) \to \mathcal{V}$ on objects by sending the string from Equation 1 to

$$F(i_n)(x_n, y_n) \otimes \cdots \otimes F(i_1)(x_1, y_1).$$

Transport via $F(f)$ and contract via composition in the $\mathcal{V}$-enriched category $F(i)$.

Colimits in sSet-Cat

In [Lur08, p. 29] each $\mathfrak{C}X$ is defined as a certain colimit in the category sSet-Cat, but colimits in sSet-Cat are notoriously difficult to understand.

— D. Dugger and D. Spivak [DS11, §1.1]

Rigidification of Simplicial Sets

Let X be a simplicial set. In [Lur08, p. 29], its **rigidification** is defined as the simplicially enriched category

$$\mathfrak{C}X := \operatorname*{colim}_{\Delta^n \to X} \mathfrak{C}\Delta^n \in \text{sSet-Cat},$$

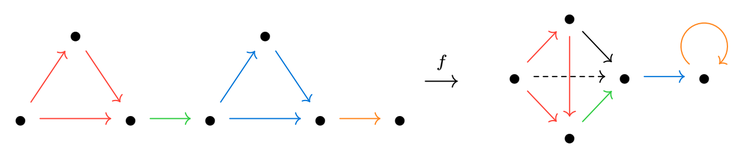
where $\mathfrak{C}\Delta^n$ denotes the simplex rigidification from [Lur08, Definition 1.1.5.1].

Necklace Construction

In [DS11, §3], Dugger and Spivak define a **necklace** as a simplicial set of the form

$$\Delta^{n_0} \vee \Delta^{n_1} \vee \cdots \vee \Delta^{n_k},$$

A **joint** is either an initial or a final vertex in some bead Δ^{n_i}. A **map of necklaces** is an endpoint-preserving simplicial map.



By [DS11, Corollary 4.4], $\mathrm{Ob}(\mathfrak{C}X) \cong X_0$ and an n-simplex in $\mathfrak{C}X(a, b)$ is an equivalence class of triples $\big[T, T \to X, \vec{T}\big]$, where

- T is a necklace,
- $T \to X_{a,b}$ is a map of simplicial sets,
- $\vec{T}$ is a flag of sets $T^0 \subseteq T^1 \subseteq \cdots \subseteq T^n$ such that T^0 contains the joints of T and T^n is contained in the set of vertices of T.

The equivalence relation is generated by

$$\big(T \to X; \vec{T}\big) \sim \big(U \to X; \vec{U}\big)$$

iff there exists a map of necklaces $f\colon T \to U$ over X with $\vec{U} = f_*\big(\vec{T}\big)$.

Colimits of Lawvere Metric Spaces

A **Lawvere metric space** is a category M enriched in the monoidal poset $([0, \infty], \geq)$ with addition as the tensor product. Objects are *points*, hom-objects $M(a, b)$ are *distances* and composition identifies with the triangle inequality

$$M(b, c) + M(a, b) \geq M(a, c).$$

Let $F\colon \mathcal{I} \to [0, \infty]\text{-Cat}$ be small and write hom-distances $d_i(x, y) := (Fi)(x, y)$. Then the hom-distance $(\mathrm{colim}\, F)(a, b)$ is the infimum over compatible sums of distances.

$$\inf\big\{d_{i_n}(x_n, y_n) + \cdots + d_{i_1}(x_1, y_1)\big\}$$